

Source-Adjusted Candidate Feature Auditing for Group-Dependent Label Corruption: Calibration, Power, Real-Data Transfer and Mitigation

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Abstract—Training labels can be corrupted in ways that depend on group membership, yet analysts are rarely told which columns are sensitive. Candidate feature auditing asks which observed feature blocks are linked to unusual label behaviour in an audit sample relative to a small trusted reference, and it must be able to abstain. We present an open reimplementaion and evaluation of a cross-fitted, source-adjusted auditor (V6) that emerged from six project versions. The test multiplies out-of-fold source and label residuals, contrasts their means between candidate groups, and applies Bonferroni correction within association blocks and Benjamini–Yekutieli correction across blocks. On 10,500 synthetic datasets, V6 kept the exact 95% upper bound of the clean alarm rate below 5% in all 20 clean conditions and abstained on 99.0% of random-label datasets, whereas the preceding permutation auditor (V5) failed two conditions and alarmed on 23.0% of datasets under covariate shift with a nonlinear label rule. A first power study (7,500 datasets) found at least 80% recall at 30% corruption but only 43–47% at 20%. On 2,200 record-disjoint semi-synthetic splits of COMPAS and German Credit, V6 recovered injected race-linked corruption in 59.5% of COMPAS splits, where V5 could not test race at all. After a V6 flag, audit-guided reweighting reduced the corrupted group’s prediction bias from 14.3 to 1.6 percentage points. Calibration on real data and sensitivity to mild corruption remain open.

Index Terms—algorithmic auditing, label bias, cross-fitting, conditional independence testing, false discovery rate, fairness

I. INTRODUCTION

Labels are often produced by processes that treat groups differently. Re-arrest records used as recidivism labels depend on where policing is concentrated [1], [2], and recorded credit outcomes reflect earlier lending decisions. A model trained on such labels reproduces the pattern even when the protected attribute is removed [3]. Most corrective methods assume that the affected group is specified in advance [4]–[7]. An analyst may instead receive a table with many columns and no statement of which ones are sensitive.

Candidate feature auditing addresses a narrower, earlier question. Given an *audit* sample whose labels are under review and a smaller *trusted* sample with reliable labels, which observed feature blocks, if any, are linked to a label difference between the two sources that the other features do not explain? Three tasks must be kept apart. *Ranking* orders features by a score. *Detection* decides, with error control, whether any feature warrants review, and may abstain. *Mitigation* changes

TABLE I
DEVELOPMENT HISTORY OF THE AUDITOR [8]

Ver.	Main change	Finding in the archived study
V1	local neighbour disparity, top-3 ranking	selected three features in every null control
V2	empirical null, BH, gated reweighting	leakage and adaptive draws; claims withdrawn
V3	trusted calibration sample, blocks, coherent score	9/9 matched clean cells passed; no 20% cell reached 80% recall
V4	trusted-sample allocation, BY vs. Holm	recall up in 18/18 cells; primary failed calibration
V5	frozen validation, 21 conditions	18 matched passed; mean-shift controls failed
V6	cross-fitted residual test	24/25 passed (upper bound 5.39%); power not run

the data and must be evaluated on its own. Ranking alone is insufficient, since pure noise also has a top-ranked feature.

Our project developed this auditor over six versions [8], [9] (Table I). The first version ranked features without a calibrated null; the second added one but was later found to leak information. Versions 3–5 compared audit labels with an independent trusted calibration sample using conditional source permutations. They controlled alarms in matched populations but alarmed under clean covariate shift. V6 replaced the permutation null with a cross-fitted residual test that models source membership. Its archived study passed 24 of 25 calibration conditions; because the protocol required every condition to pass, corruption power was never measured.

This paper makes four contributions. (i) An open, tested reimplementaion of the V6 auditor and the V5 comparator from the written method, with a per-group effect size based on partialling out [10], [11]. (ii) A paired calibration study on 10,500 datasets in 21 conditions, including covariate shift combined with a nonlinear label rule. (iii) The first power study of V6 (7,500 datasets), showing where the 80% recall target is and is not met. (iv) Semi-synthetic real-data transfer on COMPAS and German Credit with record-disjoint splits, followed by an evaluation of audit-guided mitigation on clean held-out labels. Section VII states what the evidence does not establish.

II. RELATED WORK

Label noise. Classical work models noise as instance- or class-conditional and aims at robust learning or data clean-

ing [12], [13]. Our target is different: a group-linked difference between two sources, reported as a review signal rather than corrected automatically.

Label bias and fairness. Jiang and Nachum reweight examples to correct biased labels [6], and Blum and Stangl analyse when fairness constraints recover from biased labels [7]. Both assume that the group is known. Reweighting and disparate-impact repair also require the protected attribute [4], [5]. Methods for fairness without demographics, such as distributionally robust optimisation [14] and adversarially reweighted learning [15], change the training objective. They do not decide, with error control, whether a specific feature is implicated. We use group criteria such as equalised odds [16] only to evaluate downstream models; the project’s first version drew on individual fairness [17].

Testing. The V6 statistic is closely related to the generalised covariance measure of Shah and Peters [18], which tests conditional independence through the mean of products of regression residuals. It also follows the cross-fitting and partialling-out ideas of double machine learning [10], [11]. Instead of a single global test of $S \perp Y \mid X$, V6 contrasts residual-product means between groups of a candidate feature, so a source difference shared by all groups does not produce a flag on its own. Multiplicity across blocks is handled by the Benjamini–Yekutieli (BY) procedure [19], which is valid under arbitrary dependence. Holm [20] and Benjamini–Hochberg [21] appear in earlier versions. Covariate shift without a change in $P(Y \mid X)$ [22] is exactly the setting in which naive source comparisons raise false alarms.

III. PROBLEM SETTING

Let X denote the observed features, $Y \in \{0, 1\}$ the observed label and $S \in \{0, 1\}$ the source, with $S = 0$ for trusted records (clean labels) and $S = 1$ for audit records. Each candidate feature j induces groups (quantile bins or category levels) that are fixed before any outcome is examined. The auditor is never told which columns are sensitive.

Clean null. The audit labels are uncorrupted if $P(Y=1 \mid X, S=1) = P(Y=1 \mid X, S=0)$. The covariate distribution may still differ between the sources (population shift). *Group-dependent corruption.* In the audit source, a fraction c of eligible labels inside a group g^* are flipped, so the conditional label distribution differs between sources only within g^* .

Output. The auditor returns flagged feature blocks with effect sizes and adjusted p -values, or the abstention “No statistically supported candidate detected.” A flag is a prompt for human review, not proof of discrimination, and an abstention does not certify the data.

Gates. The project protocol [8] defines three gates. (G1) In every clean condition, the exact two-sided 95% Clopper–Pearson [23] upper bound on the alarm rate (the probability of any flag) must lie below 5%. (G2) Under random labels, the lower bound on the abstention rate must be at least 90%. (G3) Recall of the corrupted block must reach at least 80% in each 20–30% corruption cell, and is interpreted only after G1 and G2 pass.

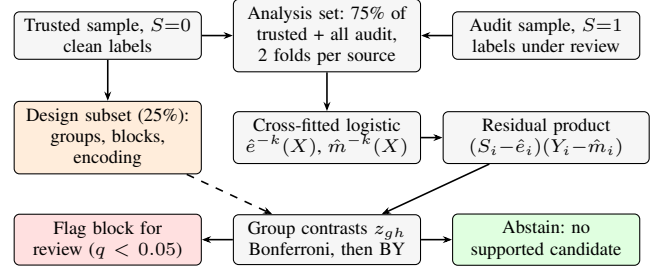


Fig. 1. The V6 audit. The design subset fixes groups and blocks (dashed) before any analysis label is used; every analysis record is predicted by models that did not train on it.

IV. METHOD

A. Design geometry

A random 25% of the trusted records forms a *design subset*. Its labels are used only to fix three things: candidate groups, feature blocks and preprocessing. Numeric features with more than six distinct values are cut into quartile bins. Other features use each level that holds at least 3% of design records, with the remaining levels merged. Preprocessing uses median imputation, standardisation and one-hot encoding. Association is $|\rho_{\text{Spearman}}|$ for numeric pairs and Cramér’s V between group codes otherwise. Blocks are the connected components of the graph with an edge wherever association is at least 0.60. The archived study set this threshold after showing that a binary proxy flipped for 10% of records at 25% prevalence has population association ≈ 0.756 , which an earlier 0.80 threshold missed [8]. Our implementation reproduces that value.

B. Cross-fitted residual contrasts

The remaining trusted records and all audit records form the analysis set, which is split into two folds within each source. For fold k , ℓ_2 -regularised logistic regressions fitted on the other fold give $\hat{m}^{-k}(x) \approx P(Y=1 \mid x)$ and $\hat{e}^{-k}(x) \approx P(S=1 \mid x)$ from all features. For record i in fold $k(i)$,

$$r_i = (S_i - \hat{e}^{-k(i)}(X_i))(Y_i - \hat{m}^{-k(i)}(X_i)). \quad (1)$$

Under the clean null with correct nuisance functions, $E[r \mid X] = 0$, so the mean of r is zero within every group [18]. For groups g and h of feature j ,

$$z_{gh} = \frac{\bar{r}_g - \bar{r}_h}{\sqrt{s_g^2/n_g + s_h^2/n_h}}, \quad (2)$$

with a two-sided normal p -value. A group is *supported* if it has at least 20 analysis records from each source, including at least 10 per source in each fold; each source also needs at least 100 analysis records. Unsupported contrasts keep $p = 1$ and remain in the family. The p -value of block b is $\min(1, M_b \min p)$ over its M_b contrasts (Bonferroni). BY is then applied across blocks, and supported blocks with $q < 0.05$ are flagged (Fig. 1, Algorithm 1).

Algorithm 1 V6 candidate feature audit

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1: Draw design subset  $D$  (25% of trusted); fit groups, blocks (assoc.  $\geq 0.60$ )
   and encoding on  $D$  only
2: Analysis set  $A \leftarrow (\text{trusted} \setminus D) \cup \text{audit}$ ; split each source into folds 1, 2
3: for  $k \in \{1, 2\}$  do
4:   Fit  $\hat{m}^{-k}, \hat{e}^{-k}$  (logistic) on fold  $3 - k$ ; predict fold  $k$ 
5: end for
6:  $r_i \leftarrow (S_i - \hat{e}_i)(Y_i - \hat{m}_i)$  for all  $i \in A$ 
7: for each feature  $j$  and group pair  $(g, h)$  do
8:    $p_{gh} \leftarrow$  normal test of (2) if both groups supported, else 1
9: end for
10:  $p_b \leftarrow \min(1, M_b \min_{(g,h) \in b} p_{gh})$ ;  $q \leftarrow \text{BY}(p_b)$ 
11: return blocks with  $q_b < 0.05$ , or “No statistically supported candidate
    detected”
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C. Per-group effect size

For reporting, we use the partialling-out estimate [10]

$$\hat{\theta}_g = \frac{\sum_{i \in g} r_i}{\sum_{i \in g} (S_i - \hat{e}_i)^2}, \quad (3)$$

the coefficient from regressing the label residual on the source residual within group g . It estimates the audit-minus-trusted difference in label probability inside g after both nuisance models have used X ; we report it with a delta-method interval. A source effect common to all groups, such as a different base rate, is not group-specific and cancels in (2).

D. Comparator (V5)

V5 [8] fits, on design data, a risk model that excludes the block and forms four risk strata. Within each stratum s and group g it computes $d(s, g)$, the audit-minus-calibration positive rate, and the coherent score $T = \max_g \sum_s w(s) d(s, g) - \min_g \sum_s w(s) d(s, g)$ with harmonic-support weights. The block maximum is calibrated with $B = 1,999$ hypergeometric source permutations within (block pattern $\times$ stratum) cells. A stratum is eligible only if *every* group has at least 10 records from each source.

E. Audit-guided mitigation

Mitigation runs only after a flag. For the lead feature of the flagged block, a logistic model trained on trusted records predicts clean probabilities. Within each group g of the audit records, let E_g be the mean predicted probability and O_g the observed positive rate. Positives receive weight E_g/O_g and negatives $(1 - E_g)/(1 - O_g)$. Weights are clipped to $[0.5, 2]$ and normalised, and the intervention is rejected if Kish’s effective sample size [24] falls below 70% of n .

V. EXPERIMENTAL DESIGN

A. Synthetic family

Five legitimate predictors $x_1, \dots, x_5 \sim N(0, 1)$ enter the logit with coefficients $(0.8, -0.6, 0.5, 0.4, -0.3)$ and intercept -0.3 . A binary group a has prevalence 25% and a legitimate effect of 0.3; a proxy p equals a flipped with probability 10%; and five $N(0, 1)$ columns carry no signal. By default there are 5,000 audit and 1,667 trusted records. Corruption relabels audit records with $a = 1$ and clean label 1 as 0 with probability c .

A shift adds δ to the audit mean of x_1 . The nonlinear rule adds $0.6(x_1^2 - 1) + 0.5x_2x_3$ to the logit.

The calibration study covers 20 clean conditions. They vary proxy noise (0, 30, 50%), audit size (1,000; 10,000), trusted size (833; 3,333), prevalence (10, 50%), 20 noise columns, nonlinear labels, 30% missingness and correlated nuisance columns ($\rho = 0.7$). They also include shifts $\delta \in \{0.75, 1.50\}$ and four combinations of $\delta = 1.50$ with nonlinear labels, correlated nuisance columns, 20 noise columns or 10% prevalence. A random-label control completes the set, with 500 datasets per condition (10,500 in total). The power study crosses $c \in \{10, 20, 30, 40\}\%$ with proxy noise $\{0, 10, 30\}\%$ and adds three cells: $c = 30\%$ with $\delta = 1.5$, and $c \in \{20, 30\}\%$ with 10,000 audit records (7,500 datasets). Both methods run on every dataset, and paired differences are tested with exact McNemar tests [25].

B. Real-data transfer

We use COMPAS (6,172 defendants after ProPublica’s filtering [1]; label two-year re-arrest; nine features) and German Credit (1,000 applicants [26]; 20 features; label bad risk). Each split first draws a 25% clean test set. Of the remainder, 30% becomes trusted, and 70% of what is left is sampled as audit, either uniformly or with probability $\propto \exp(\gamma z_{\text{age}})$ for a clean shift with $\gamma \in \{1.0, 1.5\}$. Corruption relabels audit negatives as positives with probability c in the target group: race = African-American in COMPAS, and female (divorced, separated or married) in German Credit. We draw 200 splits per condition (2,200 in total).

C. Mitigation evaluation and protocol

In corrupted splits where V6 flagged, we compare five strategies: training on the audit labels (baseline), audit-guided reweighting, dropping the flagged block, training on trusted records only, and an oracle trained on the clean audit labels. Each is fitted with logistic regression (LR) and a 100-tree random forest (RF) [27]. On clean test labels we report AUC, the equalised-odds (EO) gap, $\max(|\Delta \text{TPR}|, |\Delta \text{FPR}|)$ between the target group and the rest [16], and the group prediction bias, defined as the mean predicted probability minus the clean positive rate in the target group.

The code, gates and condition list were fixed before the main runs. A 100-repetition pilot on four conditions preceded them, after which two trusted-size conditions were added; no auditor code was changed. This is therefore a development and validation study by the method’s developers, not a locked confirmation study.

D. Implementation and interactive tool

The auditor is a Python package built on NumPy, pandas, SciPy and scikit-learn [27]. Its 25 unit tests include reproductions of the interval values quoted in the project protocol, such as the 5.39% upper bound for 17 alarms in 500 runs and the 0.756 proxy association. A web application exposes the same code in four views. A *synthetic lab* runs clean, corrupted and shifted scenarios with V6 and V5 side by side and reveals the

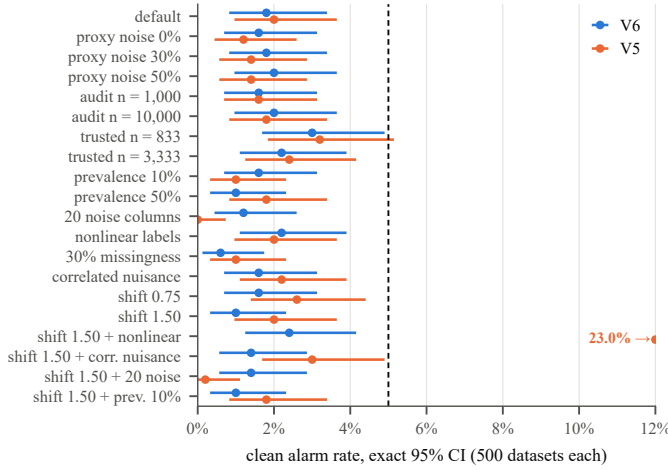


Fig. 2. Clean alarm rate per condition (paired, 500 datasets each) with exact 95% intervals. The dashed line marks the 5% gate on the upper bound; the arrow gives the V5 rate beyond the axis.

ground truth afterwards. A *real-data audit* accepts COMPAS, German Credit or an uploaded CSV, injects corruption or shift, and then trains the downstream models. A *calibration* view runs live Monte-Carlo alarm studies with exact bounds, and a *method* view states the limits of a flag. Every audit exports a JSON report with all contrasts, supports and adjusted p -values. The implementation, study scripts and per-run records are available from the authors.

VI. RESULTS

A. Calibration

Fig. 2 and Table II summarise the study. V6 passed G1 in all 20 clean conditions, with alarm rates of 0.6–3.0%. Its largest upper bound, 4.90%, came with only 833 trusted records (15/500 alarms). It abstained on 495 of 500 random-label datasets (lower bound 97.7%), passing G2. V5 passed 18 of 20 conditions. It failed with 833 trusted records (upper bound 5.14%) and under shift with nonlinear labels, where it alarmed on 115 of 500 datasets (23.0%, CI 19.4–26.9%) against 12 of 500 for V6. On the same datasets, 110 runs alarmed under V5 alone and 7 under V6 alone (exact McNemar $p < 10^{-24}$). In our generator, V5 passed both linear shift conditions.

V5 raised no alarms with 20 noise columns, but this is not evidence of good calibration. With $B = 1,999$, the smallest attainable p is $1/2000$, which exceeds the BY rank-one cutoff for 26 blocks, $0.05/(26 \sum_{i \leq 26} i^{-1}) \approx 0.000499$. V5 therefore cannot flag an isolated block in that setting [8]. The median run time per dataset was 0.19 s for V6 and 0.38 s for V5.

B. Power

At $n = 5,000$ (Fig. 3), V6 recall was 3.0–5.4% at 10% corruption, 42.6–46.8% at 20%, 88.0–91.6% at 30% and at least 99.6% at 40%. G3 is therefore met at 30% but not at 20%; with 10,000 audit records, recall at 20% rose to 83.8%. The paired comparisons were mixed. V6 was better with a noisy proxy (at $c = 20\%$ and 30% noise, 59 vs. 7 discordant runs,

TABLE II
RESULTS AGAINST THE PROTOCOL GATES

Endpoint	V6	V5
G1: clean conditions passed (synthetic)	20/20	18/20
largest upper bound	4.90%	26.9%
G2: random-label abstention	99.0%	99.2%
lower bound	97.7%	98.0%
G3: cells $\geq 80\%$ recall, $c \in \{20, 30\}\%$, $n=5k$	3/6	3/6
recall at $c = 20\%$, $n = 10k$	83.8%	81.2%
COMPAS clean alarms (200 splits)	2.5%	0.5%
COMPAS recall of race, $c = 30\%$	59.5%	0.5%

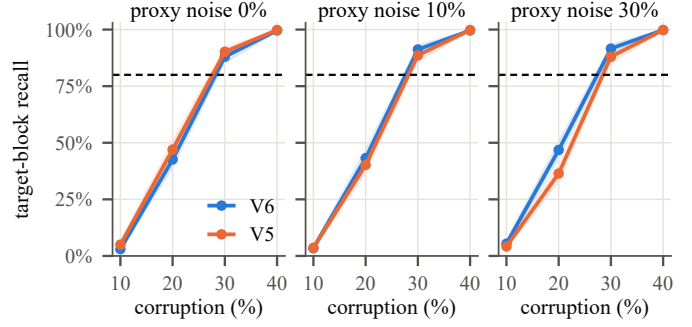


Fig. 3. Recall of the corrupted block at $n = 5,000$ (500 datasets per point, shaded 95% intervals). The dashed line marks the 80% target.

$p < 10^{-10}$). V5 was better at $c = 20\%$ with an exact proxy (39 vs. 17, $p = 0.005$) and under shift (93.4% vs. 81.6%; 62 vs. 3, $p < 10^{-14}$). Off-target selections stayed at or below 5.8%, except at 30% proxy noise. There the proxy’s association with a is about 0.35, below the 0.60 threshold, so p forms its own block. Selecting it is a genuine proxy signal, but we count it as off-target (up to 24.6% at $c = 40\%$).

C. Effect-size accuracy

For the corrupted group $a = 1$, the realised gap is the mean of $Y - Y^*$ over audit records with $a = 1$. Over 300 datasets per corruption level (Table III), $\hat{\theta}_g$ from (3) was essentially unbiased, and its 95% interval covered the realised gap in 95.0–97.0% of datasets. The flag can therefore be read with a calibrated magnitude: at $c = 30\%$, “audit labels in $a=1$ are about 15 percentage points lower than trusted labels after adjustment.”

D. Real-data transfer

On COMPAS (Fig. 4), V6 recovered race in 0.5%, 16.0% and 59.5% of splits at $c = 10, 20$ and 30% , and in 59.5% at 30% combined with the age shift. V5 recall never exceeded 0.5%. In every split we inspected, V5 marked race as unsupported. The merged Asian and Native-American group (42 defendants) never reaches 10 records per source per stratum, so V5’s all-groups rule removes the whole feature from testing. V6’s pairwise support still compares the supported groups. On clean COMPAS splits, V6 alarmed in 5 of 200 (2.5%, upper bound 5.74%) and in 4 of 200 under $\gamma = 1.5$ (upper bound 5.04%), so G1 is not demonstrated on real data at this

TABLE III
RECOVERY OF THE INJECTED GAP BY $\hat{\theta}_g$ FOR $\alpha = 1$ (300 DATASETS PER ROW)

c	realised gap	mean $\hat{\theta}_g$	SD of $\hat{\theta}_g$	95% CI coverage
0%	0.000	0.001	0.027	95.3%
20%	-0.100	-0.100	0.028	95.0%
30%	-0.150	-0.147	0.028	95.3%
40%	-0.201	-0.198	0.026	97.0%

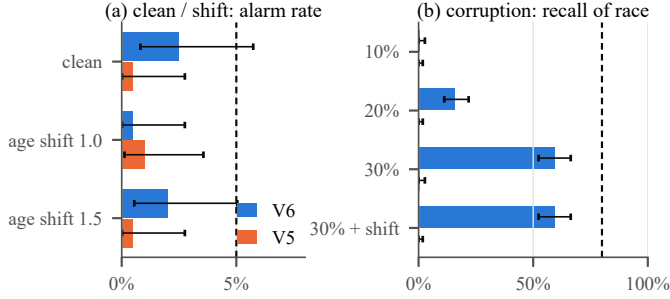


Fig. 4. COMPAS semi-synthetic transfer (200 record-disjoint splits per bar, exact 95% intervals): (a) alarm rate with no corruption, with and without an age shift; (b) recall of race at corruption rate c .

sample size; V5 alarmed in 1 of 200. On the smaller German Credit dataset, V6 recall was 4.0% at $c = 30\%$ and 35.5% at $c = 50\%$, with 1 of 200 clean alarms; V5 recall was zero.

E. Mitigation

Table IV reports the 122 COMPAS splits at $c = 30\%$ where V6 flagged. Reweighting reduced the group prediction bias from 0.143 to 0.016 with LR and from 0.114 to 0.016 with RF (oracle: 0.002 and -0.007). AUC rose by 0.016 and 0.008, and the EO gap fell from 0.652 to 0.387 (LR) and from 0.474 to 0.284 (RF). The project’s criteria (lower EO gap, AUC loss ≤ 0.02 , ESS $\geq 70\%$) were met in 97.5% (LR) and 98.4% (RF) of these splits. Dropping the flagged block gave the smallest EO gap but left a bias of 0.067. Removing race also removes the legitimate group information that the oracle retains, so the EO gap alone can reward an intervention that leaves the corruption in place. Training on trusted records only was competitive (bias ≈ 0 , LR EO gap 0.295), because the trusted sample (about 1,389 records) is large for this task. On German Credit at $c = 50\%$ (72 flagged splits), reweighting reduced the LR bias from 0.323 to 0.091 and raised AUC by 0.029.

VII. DISCUSSION AND LIMITATIONS

Calibration is necessary but not sufficient. V6 met G1 and G2 on every synthetic condition, yet it detected mild corruption poorly: 20% corruption needed 10,000 audit records, and 10% was essentially undetectable. An abstention therefore does not certify that the labels are clean. Source adjustment also has a cost. As the sources become more separable, $S - \hat{e}$ shrinks and each record carries less information, which explains V5’s higher recall under linear shift. V5 gains that recall by not adjusting, the same choice that produced its 23% alarm rate when the shift met a nonlinear label rule.

TABLE IV
MITIGATION ON COMPAS, $c = 30\%$, 122 FLAGGED SPLITS (CLEAN TEST LABELS; MEANS)

Strategy	AUC		EO gap		Group bias	
	LR	RF	LR	RF	LR	RF
Baseline (audit labels)	.706	.712	.652	.474	.143	.114
Audit-guided reweighting	.723	.720	.387	.284	.016	.016
Drop flagged block	.727	.716	.258	.223	.067	.064
Trusted records only	.724	.722	.295	.258	-.001	-.010
Oracle (clean labels)	.726	.724	.324	.266	.002	-.007

Support rules are part of the method. On real multi-level attributes, V5’s requirement that every group be represented in every stratum made race untestable. A rule that looked harmless on binary synthetic groups decided the outcome on COMPAS.

Real-data calibration is not established. Observed clean alarm rates of 2.0–2.5% on COMPAS came with upper bounds above 5%. Candidate explanations include logistic misspecification on real features, small groups and the normal approximation. Settling the question needs a larger, prespecified study.

Mitigation evidence is conditional. It covers only splits in which the auditor flagged, it relies on a label model trained on trusted records, and here training on the trusted records alone was competitive. Reweighting removed the group-level prediction bias while using every audit record. We do not claim general debiasing.

Scope. The synthetic family and conditions were chosen by the developers; nuisance models are logistic; each dataset has a single corrupted group; and our numbers come from a reimplementing of the written method, not the archived V6 code. Next steps are a locked confirmation study with prespecified real-data calibration, flexible cross-fitted nuisance models, omnibus group tests to raise power, and mixtures of corrupted groups.

VIII. CONCLUSION

Candidate feature auditing must be able to decline to flag. A cross-fitted, source-adjusted residual test held the clean alarm rate below the 5% bound in all 20 synthetic conditions, including a shift with a nonlinear label rule on which its permutation predecessor alarmed in 23% of runs. It recovered race-linked corruption on COMPAS where the predecessor could not test race at all, and its flags supported a reweighting that removed most of the induced prediction bias. The same study shows its limits: modest power below 30% corruption and calibration on real data that is not yet demonstrated. The auditor should therefore be used as a calibrated review signal, not as an automatic fairness fix.

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